

Dear Pushing / World Model Task Groups,

Here are the details for Project 2: Pushing – World Model. Please read carefully.

The core requirement of this project is that your solution must be built around a **world model approach**, your system should learn an explicit model of how the object responds to push actions, and use that model for planning and/or control. We place no restrictions on model architecture (as long as it incorporates a world model) or training procedure. You may use architectures trained from scratch, any pre-trained policy or feature extractor, and any post-training or fine-tuning method. Training data may come from public datasets, teleoperation, or simulation. Please do not use custom hardware beyond what was provided in the course kit, and treat it with care as it will be returned at the end of the course. If you are unsure whether your approach qualifies, please reach out before the evaluation.

General:

Workspace & Objects:

For all evaluations, you will be provided with:

- A rubber mat (~50 × 75 cm) to be placed on the table in front of the robot as your working space. We will draw the allowed corridor for eval 1 on it with white marker.
- A 3D-printed polyhedral object (and for Eval 3, a second unseen but similar object revealed on the day of evaluation).

The mat increases friction, so objects will tend to tip rather than slide. However, no penalty is given if you manage to slide the object instead.

The maximum score is 150 pts for the main evaluation, plus up to 50 bonus pts.

Notes:

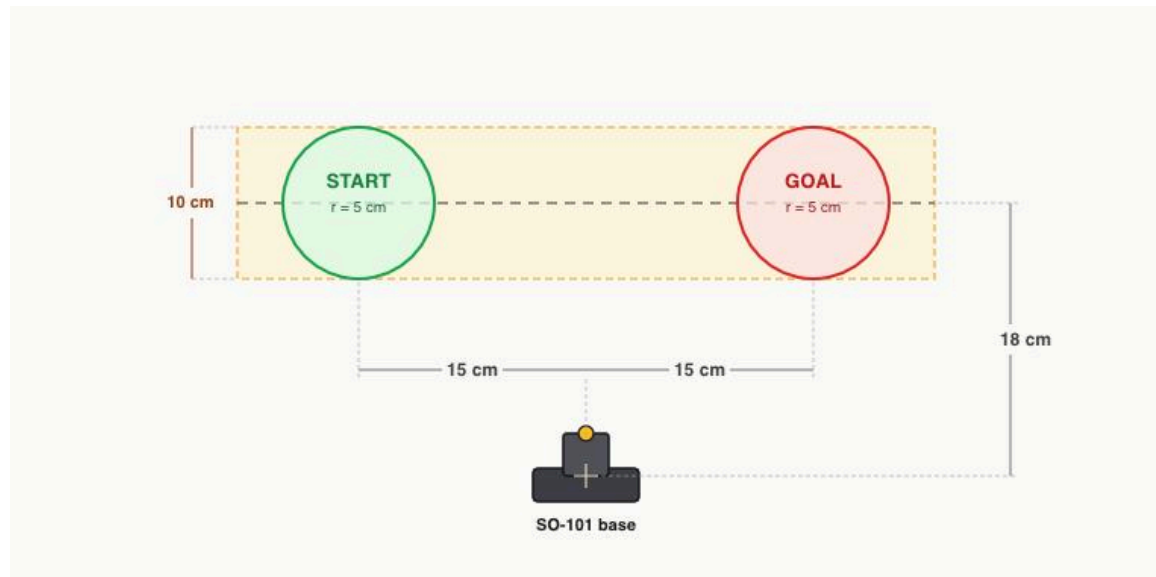
- You specifically are allowed to use a different policy/approach for every individual attempt.
- The gripper needs to stay in a fixed joint position throughout the attempt.
- We do not limit you on which World Model you use or if you implement your own. Consider taking a look at *DreamZero*, *mimic-video* (in a scaled down version) or *LeWorld*.

Deliverables:

- Demo on Demo Day in HG
- Code
- Collected Datasets

Eval 1 (50 pts) — Straight-Line Pushing

Goal: Push a small object from left to right along a straight line.



Setup

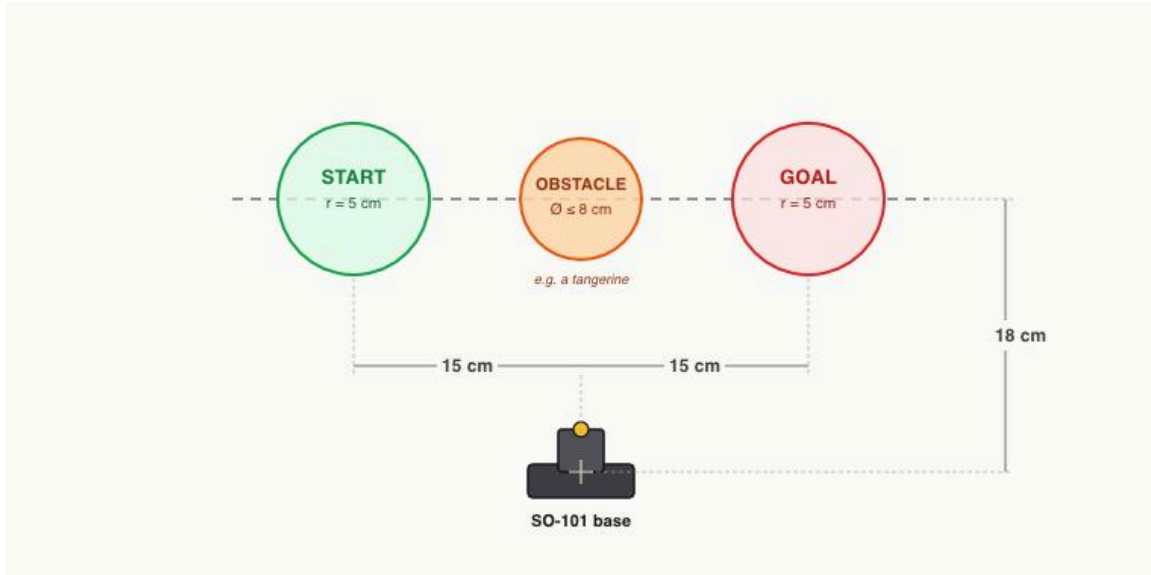
- The line is located 18 cm in front of the robot base.
- **Start region:** circle of 5 cm radius, centered 15 cm to the left of the robot base on the line.
- **Goal region:** circle of 5 cm radius, centered 15 cm to the right of the robot base on the same line.
- You may place the object anywhere within the start circle.
- The object must stay within a ± 5 cm corridor around the line (10 cm wide total) for a clean execution.
- 5 attempts, max 1 minute each. Only the highest scoring attempt counts.

Scoring

Points	Condition
50	Object reaches goal circle and stays within the corridor throughout
35	Object reaches goal circle but exits the corridor at some point
20	Object reaches the right-side position (15 cm right of base) but outside the goal circle
10	Object displaced ≥ 10 cm in any direction (no goal reached)
0	None of the above

Eval 2 (50 pts) — Pushing Around an Obstacle

Goal: Push the same object from start to goal, navigating around a static obstacle placed at the midpoint of the line.



Setup

- Start and goal regions are identical to Eval 1.
- There is no corridor requirement, the object may take any path around the obstacle.
- A static obstacle (low height, diameter $\leq 8 \text{ cm}$) is placed at the midpoint of the line. It must not be touched or moved. In practice, we will use a tangerine.
- "Passing" the obstacle means the pushed object has reached the goal-side of the obstacle (on the right side of workspace), regardless of path taken.
- Touching the obstacle counts if either the robot or the pushed object come into contact with the obstacle.
- 5 attempts, max 1 minute each. Only the highest scoring attempt counts.

Scoring

Points	Condition
50	Object reaches goal circle; obstacle never touched
40	Object reaches goal circle; obstacle was touched or moved
30	Object passes the obstacle without touching it, but misses goal circle
20	Object passes the obstacle but touched it along the way
10	Object displaced $\geq 10 \text{ cm}$ in any direction (does not reach goal-side of obstacle)
0	None of the above

Eval 3 (50 pts) — New Object

Goal: Repeat Evals 1 and 2 with a previously unseen object (similar but not identical in shape, size, or mass). The object is revealed on evaluation day.

Eval 3 is split into two subtasks:

Eval 3a (25 pts) — Straight-line push with new object. Same rules as Eval 1, scores scaled to: 25 / 17.5 / 10 / 5 / 0.

Eval 3b (25 pts) — Around-obstacle push with new object. Same rules as Eval 2, scores scaled to: 25 / 20 / 15 / 10 / 5 / 0.

5 attempts per subtask, max 1 minute each, best attempt is scored.

Bonus (50 pts) — Fewest Pushes

The bonus rewards efficient use of your world model. A push is counted every time the gripper makes contact with the object and then leaves it, regardless of whether the object moved.

The 3 groups with the fewest pushes in their successful attempt for each evaluation, provided they achieved full marks on that evaluation, each receive the entire bonus pool for that evaluation:

Evaluation	Bonus Pool	Eligibility
Eval 1	10 pts	Full success on Eval 1 required
Eval 2	10 pts	Full success on Eval 2 required
Eval 3	30 pts	Full success on Eval 3 required

Your total bonus is the sum across all pools you qualify for, capped at 50 pts.

Note: Some of the distances and limits specified above (e.g. the 30 cm total pushing range, the ± 5 cm corridor, and object size constraints) are subject to change. Final values will be confirmed on Sunday, April 26th, once we have verified that the parameters result in a fair and well-calibrated challenge across all groups.

Best regards,

Jonas, Aris and David